

NFL Analytics · Rolling Season Forecast

NFL Record Projection Model

A Rolling 2025 Season Forecast
Using Pro Football Reference Data

Project in one sentence

A rolling forecasting system that converts game win probabilities into projected final NFL team records.

Final Deliverable

544

projections · 32 teams × 17 games

Final records matter and they change every week.

01

Why projected records matter

Playoff seeding, divisional positioning, and front-office decisions all hinge on final win totals, not single-game outcomes.

02

Team strength is non-stationary

Injuries, scheme changes, and roster moves cause real week-to-week shifts. A model that doesn't update is already stale.

03

Information accumulates with time

Week 1: priors dominate. Week 12: current-season evidence dominates. The model has to span both regimes gracefully.

04

This is a rolling forecasting problem

Can't just be a single game classifier, but a season-long projection engine that re-estimates final wins before every game.

The Question

Can Pro Football Reference team-level data predict **2025 NFL final records** before every game is played?

The Deliverable

A projected final record for every team, before every 2025 game, updated as the season unfolds.

Output table size

544

projection rows

32 teams × 17 games

Built on the public record: PFR schedules.

Source

Pro Football Reference

Schedule & results pages, 2021–2025

Fields collected

· Teams & opponents

· Scores (both sides)

· Total yards

· Home / away flag

· Wins, losses, ties

· Turnovers

Why 2021–2025

The modern 17-game era. Older seasons used a 16-game schedule, so restricting to 2021–2025 keeps the record projection target consistent with the modern 17-game format.

Game counts by season	
2021	272
2022	271
2023	272
2024	272
2025	272
Total	1,359

Note · 2022 has 271 completed games because the Buffalo Bills – Cincinnati Bengals game (Damar Hamlin injury) was canceled and never replayed.

PFR ships winners and losers, not home and away.

The raw table format

PFR lists every game with a **Winner** and a **Loser** column which is convenient for box scores, but useless for modeling because I need to know *which side was at home*.

The decoding rule

"@" in the column → Winner played AWAY

The loser was at home.

blank → Winner played HOME

The loser was the visitor.

From there, I can derive clean `home_team`, `away_team`, `home_score`, `away_score` columns.

2024 cleaning — row-count funnel

Raw PFR table

303 rows

filter to regular season · drop playoff & header rows

Regular-season games

272 rows

decode "@" rule · rename columns

Clean game-level table

`home_team` · `away_team` · `home_score` ·
`away_score`

Single function over five seasons: 1,359 games.

From script to reusable function

The 2024 cleaning logic became a single reusable function.

For each season I saved the PFR HTML locally, then call the same function. It filters to the regular season, renames columns, and decodes the home / away rule. Saved HTML makes the pipeline reproducible without re-hitting PFR.

- 1 Read saved PFR HTML for the season
- 2 Filter rows to regular season only
- 3 Rename winner/loser → home/away using "@"
- 4 Concatenate all seasons into one game-level dataset

Combined game-level dataset

2021		272
2022		271*
2023		272
2024		272
2025		272

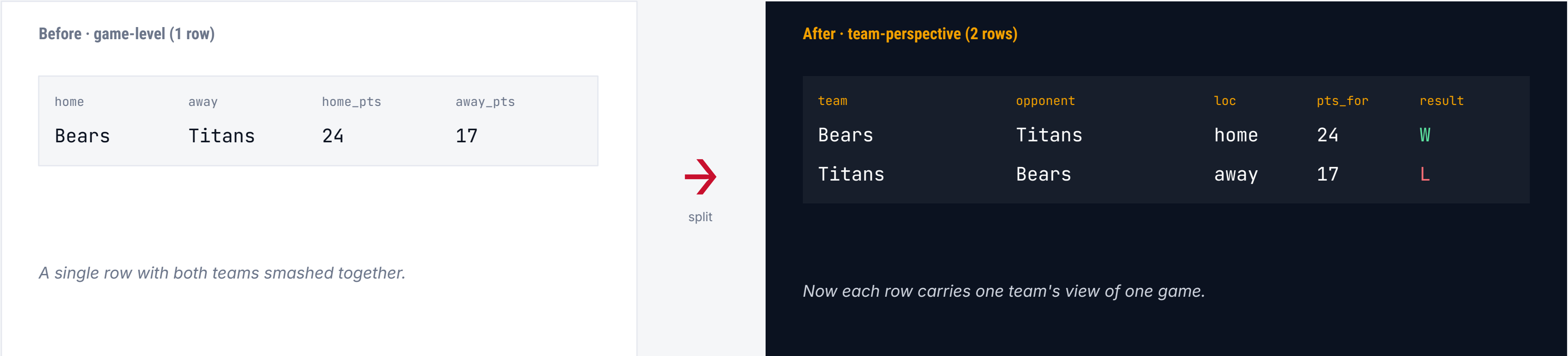
Total games, 2021–2025

1,359

* 2022 has 271 because the Bills – Bengals game was canceled and never replayed as mentioned previously.

One game $\neq$ one row. We need **a team's view**.

A game has two teams, each with its own pre-game context. To model from a team's perspective, every game becomes *two rows*: one from each side.



Team-game rows

2,718

Validation checks

- ✓ Every game appears **exactly twice**
- ✓ Margins **sum to zero** per game
- ✓ Each row is **W / L / T**

Every feature uses **only information available before the game.**

Anti-leakage principle

For each team-game row, all features are computed using stats from **prior games only**, never the game being predicted. Cumulative stats are shifted by one game so the current game result is never included.

Team's own pre-game stats

- team_wins_entering
- team_losses_entering
- team_ties_entering
- team_win_pct_entering
- team_pf_pg_entering
- team_pa_pg_entering
- team_pd_pg_entering

Record + scoring rates entering the game.

Opponent mirror set

- opp_wins_entering
- opp_losses_entering
- opp_ties_entering
- opp_win_pct_entering
- opp_pf_pg_entering
- opp_pa_pg_entering
- opp_pd_pg_entering

Same set, computed for the other side.

Difference features

- win_pct_diff_entering
- pf_pg_diff_entering
- pa_pg_diff_entering
- pd_pg_diff_entering

team - opponent, computed per row

Where most of the model's predictive signal lives.

Before Week 1, the model needs a memory.

The cold-start problem

Every team enters Week 1 with no current-season data.

Rolling stats are undefined when no games have been played. Without a fallback, the model has nothing to say in early weeks which is exactly when projections are most needed.

Solution: add last season's performance as a prior.

"Before current-season data exists, the model leans on what the team looked like last year."

Previous-season features

01 Previous win percentage

prev_win_pct

02 Previous points for per game

prev_pf_pg

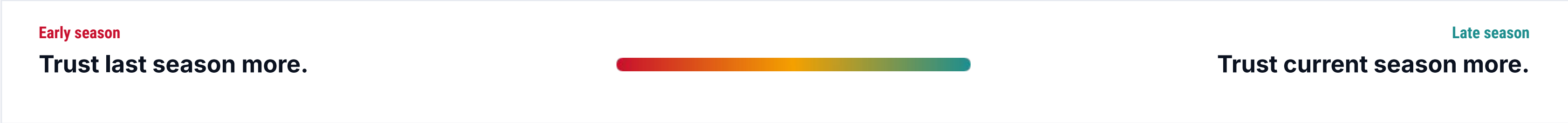
03 Previous points against per game

prev_pa_pg

04 Previous point differential per game

prev_pd_pg

Blended strength: a prior that fades as evidence arrives.



The formula

```
weight that scales with games played:  
  
current_weight = games_played_entering /  
  (games_played_entering + 4)  
  
-----  
  
blended point differential per game:  
  
blended_pd_pg = current_weight · current_pd_pg  
                + (1 - current_weight) · prev_pd_pg
```

Current_weight by games played		
Games	Weight	Lean toward
0	0.00	100% prior
4	0.50	50 / 50
8	0.67	two-thirds current
16	0.80	mostly current

Do the engineered features actually predict winning?

Questions I asked the data

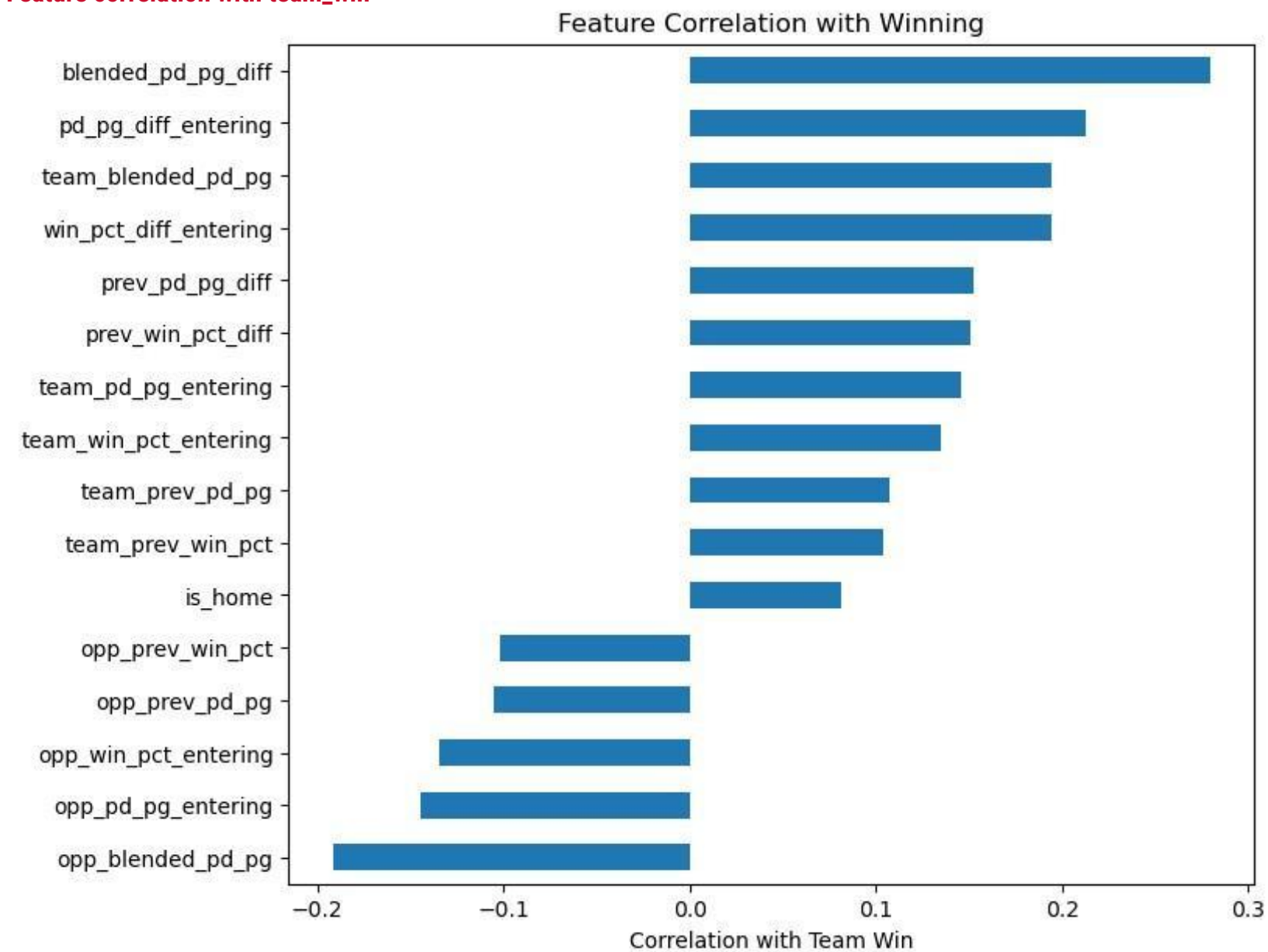
Do home teams win more often?
Yes. ~54% home, ~46% away across 2021–25.

Does point-differential advantage track win rate?
Yes. Monotonic, near-linear relationship.

Does previous-season strength still matter?
Yes. Smaller signal, but real and persistent.

Does **blended strength show signal?**
Blended strength showed one of the strongest single-feature signals.

Feature correlation with team_win



Start simple with **logistic regression**, time-based split.

Why logistic regression

- **Interpretable.** Each coefficient maps cleanly to a feature.
- **Probabilistic.** Outputs interpretable win probabilities.
- **Right shape.** Win / loss is binary. This is the textbook fit.
- **Probabilities matter** downstream. Projections sum them.

Target & setup

Target: **team_win** · ties removed for the binary model.

Time-based split, no future leakage

Train

2021 - 2023



Validate

2024



Test

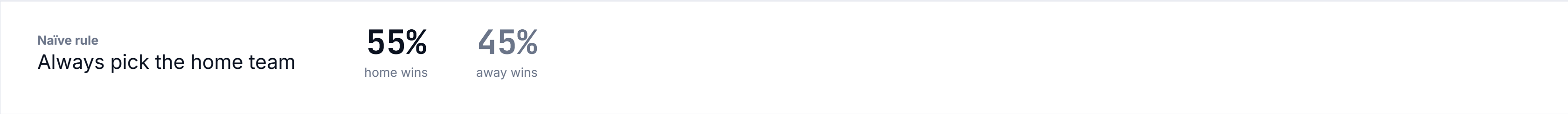
2025



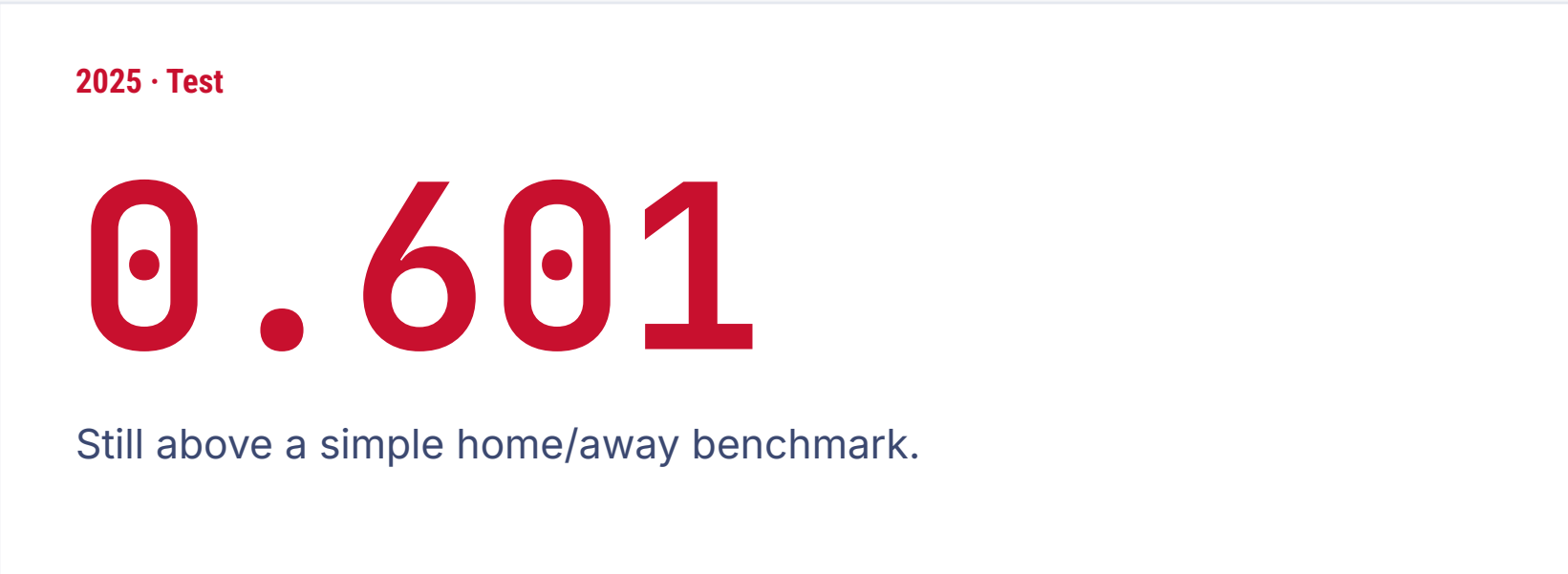
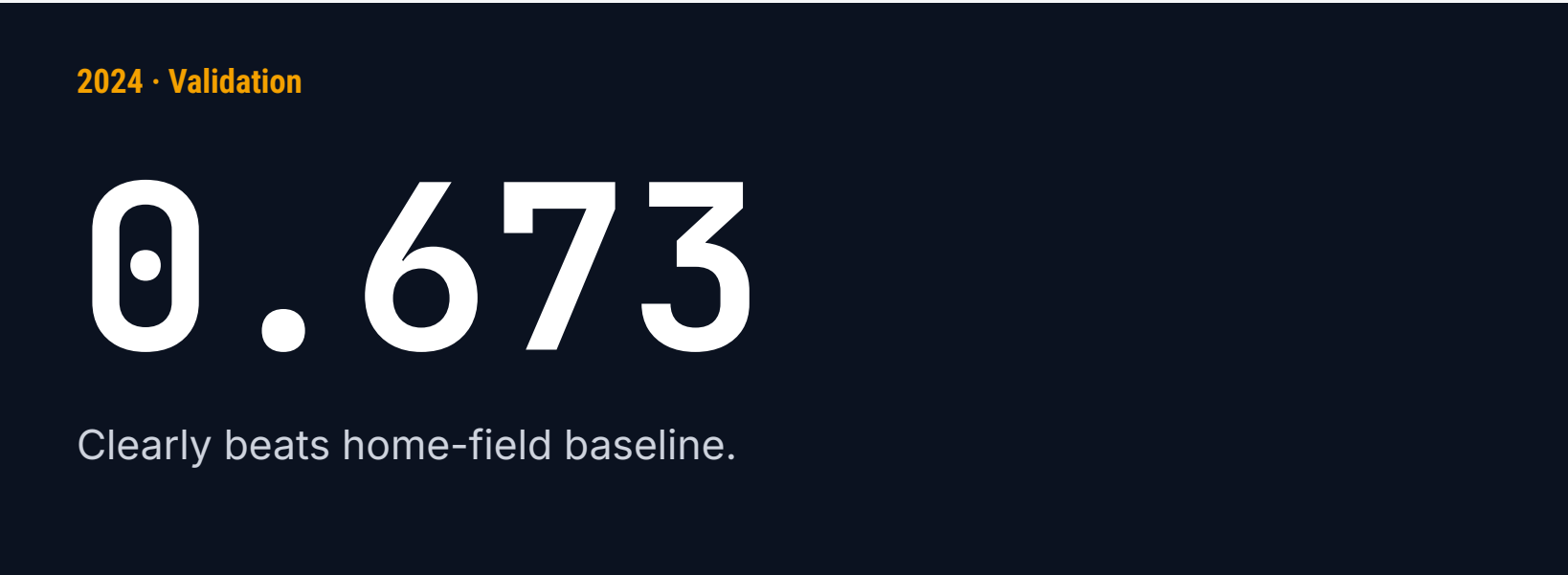
Train on past, validate on the next season, test on the one after. Random splits would leak future games into training.

Beats the baseline, and probability quality is what matters next.

Baseline



Logistic regression: first-pass results



NFL prediction is irreducibly noisy. The model beats baseline meaningfully but the next test is *probability quality*, because record projections sum these probabilities.

I tested three families, and the simplest one won.

Models compared · selection = best 2024 validation log loss

Candidate model	Validation log loss	Accuracy	Complexity	Verdict
Baseline logistic default C	0.6312	0.6728	low	<i>strong reference</i>
Tuned logistic C = 10	0.6251	0.6728	low	selected
Random forest	0.6280	0.6654	high	<i>no clear gain</i>
Gradient boosting	0.6362	0.6673	high	<i>no clear gain</i>

Why tuned logistic (C = 10) won

Best probability quality.

Lowest 2024 validation log loss across all candidates.

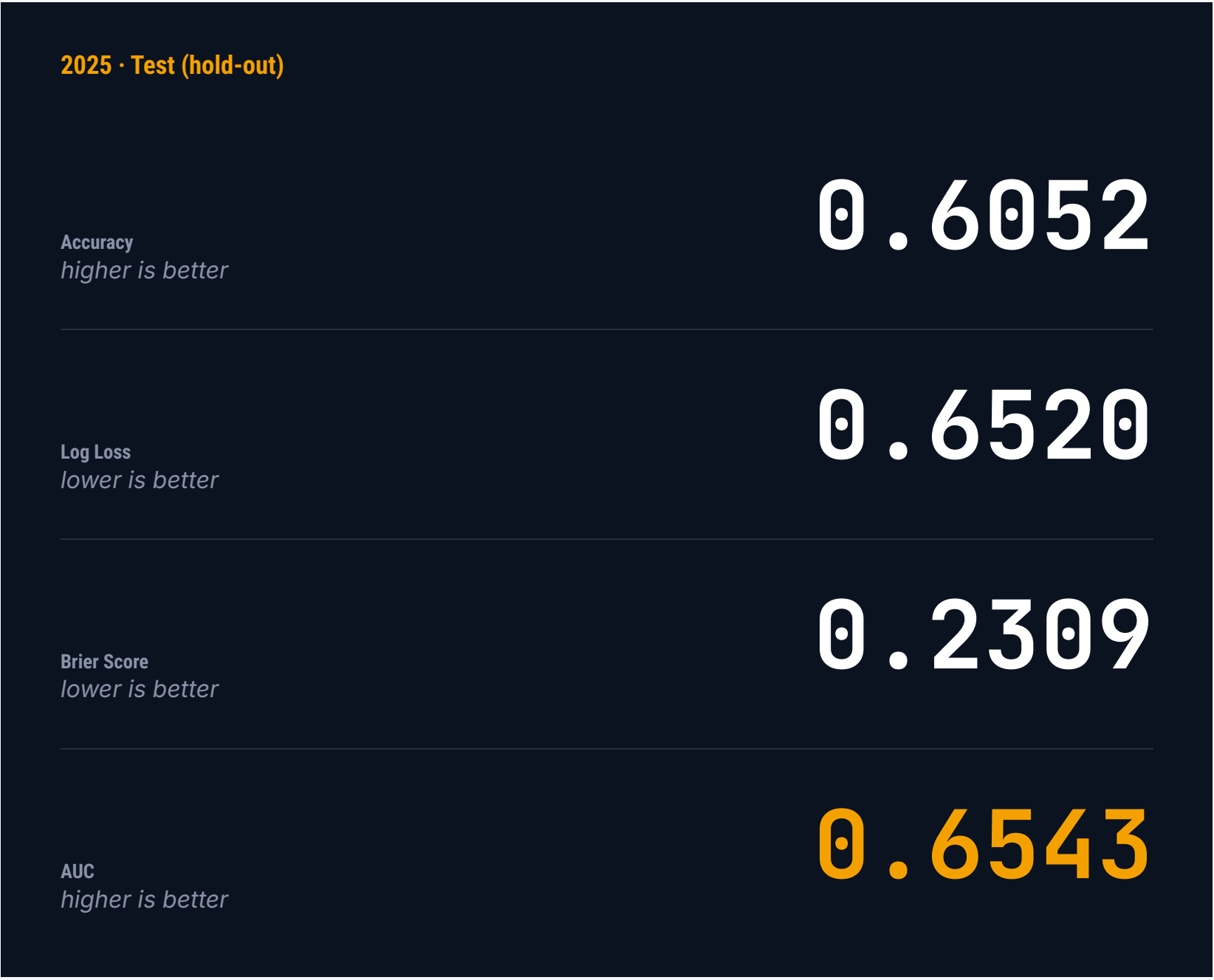
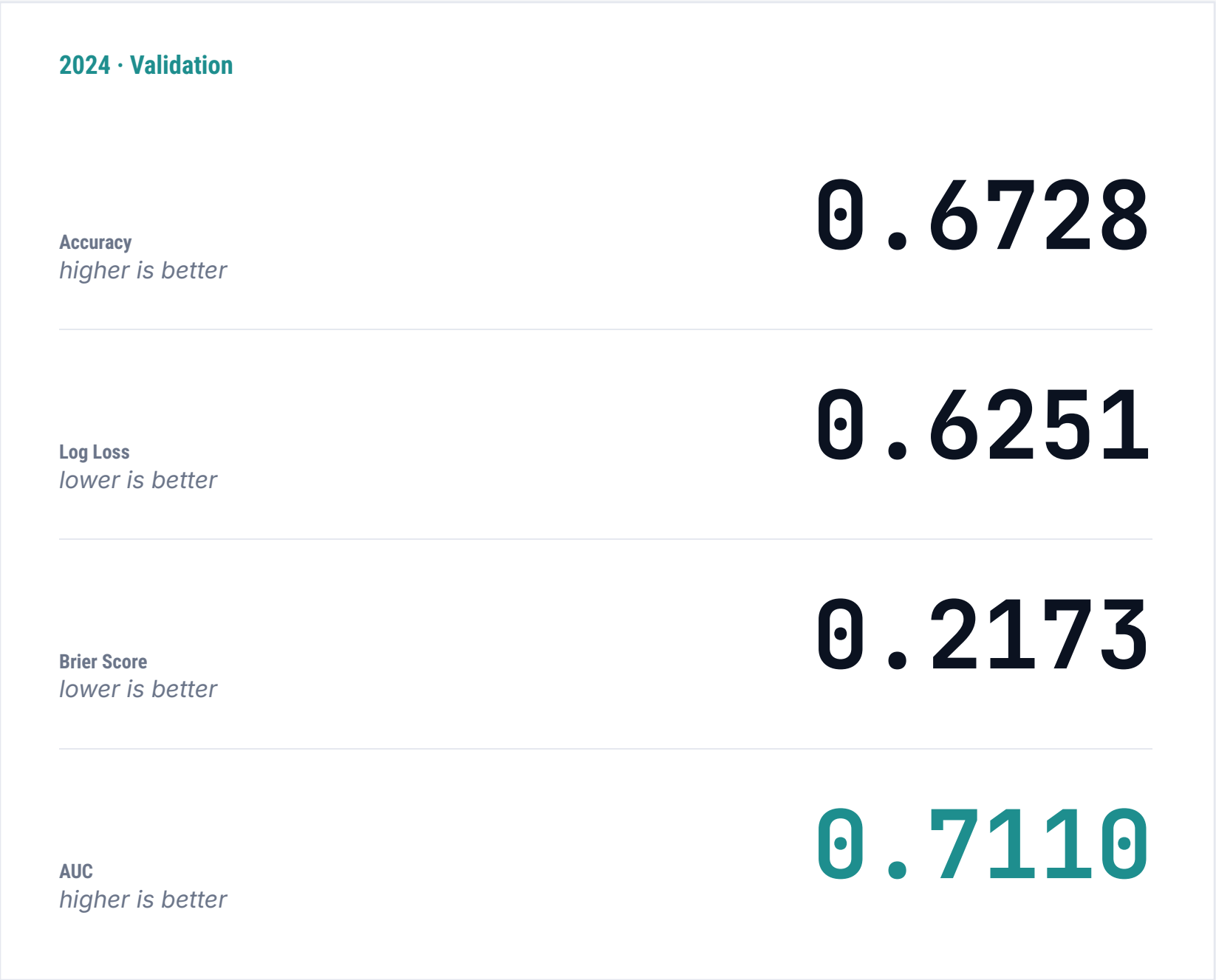
Tree models didn't deliver.

Random forest & gradient boosting offered no clear gain on this feature set.

Interpretability survives.

Coefficients still readable. One can explain *why* a team is favored.

Four metrics, two seasons: a complete report card.



Honest read · 2025 lower than 2024 but well above baseline: real signal with PFR-only data.

The core deliverable: **a record, not a game pick.**

Projection formula

Projected Final Wins = Current Wins + 0.5 × Current Ties + Σ Remaining Win Probabilities

Ties count as half a win, which matches NFL standings convention.

For each team, before each game

01

Count current record

Wins + ties × 0.5 from games already played.

02

Identify remaining schedule

Every game not yet played from this row's perspective.

03

Predict remaining games

Use the tuned logistic win probability for each remaining matchup.

04

Sum & report projected wins

Add current record + sum of predicted win probabilities.

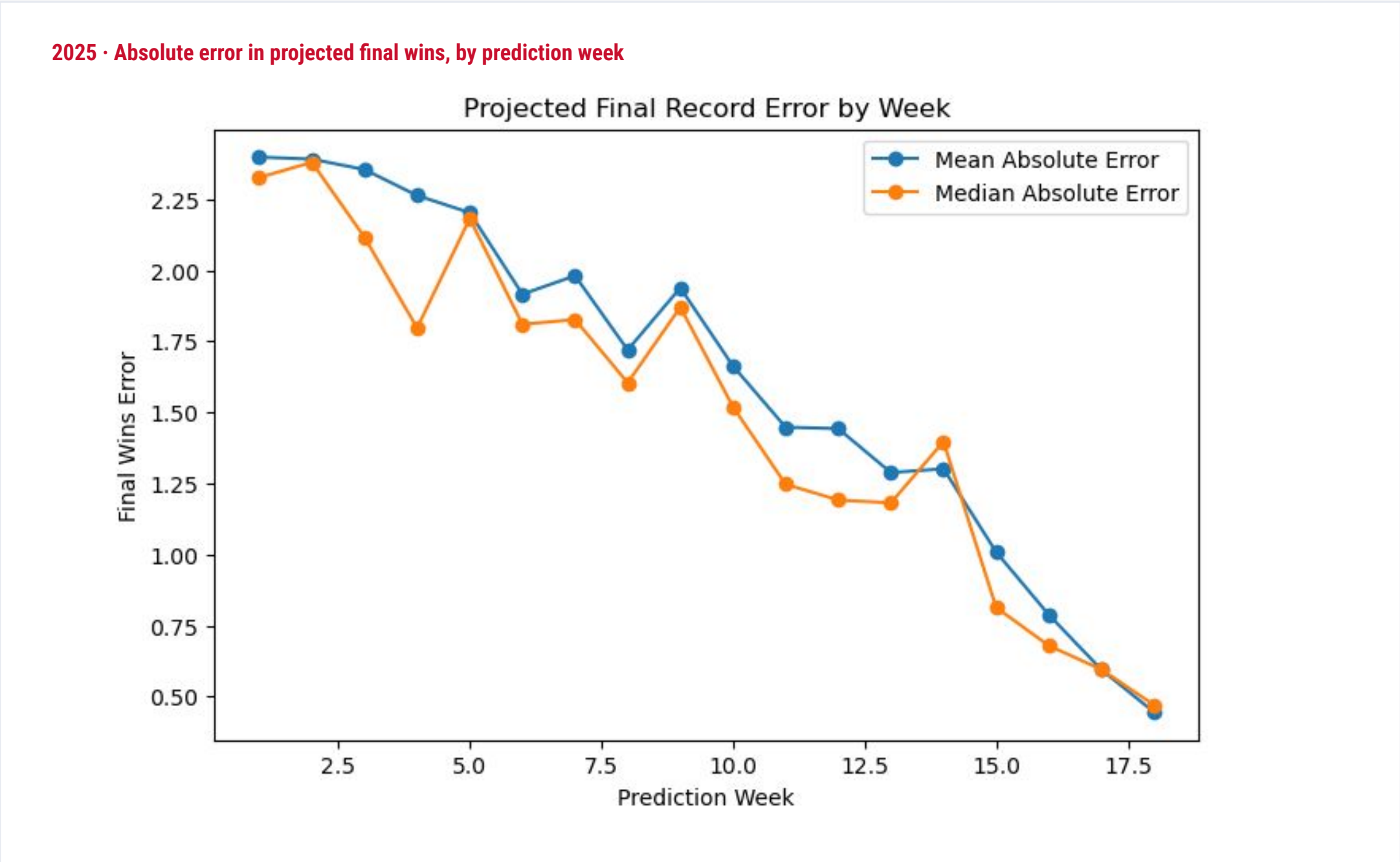
Output file

record_projections_2025.csv

Final shape

544 rows × 26 columns

Projections sharpen as the season unfolds.



What the metric measures

Absolute error = $|\text{projected} - \text{actual}|$
in win-equivalent units · ties count as 0.5

Week 1 → Week 18

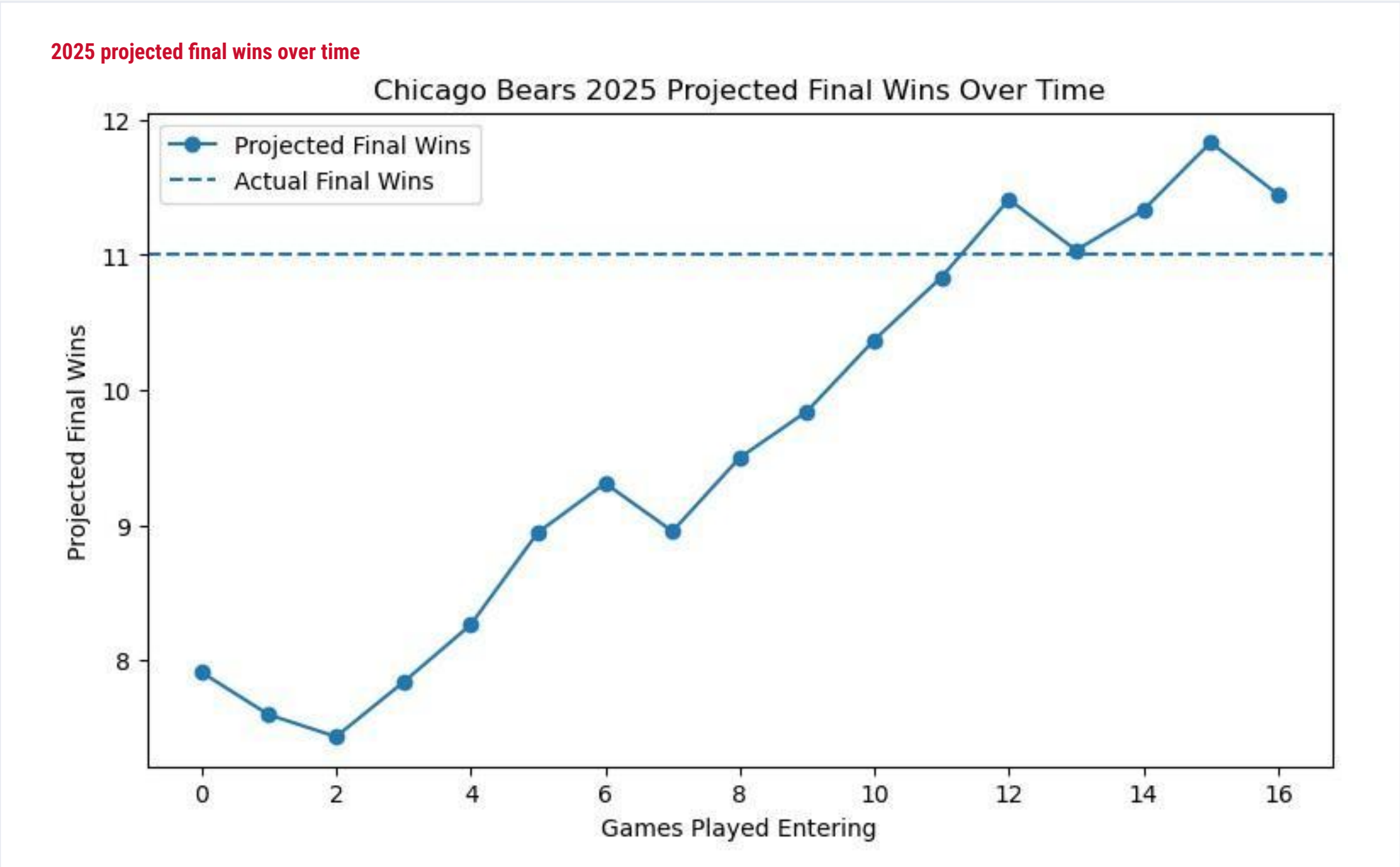
~2.4 → ~0.5

Mean absolute error in final wins.

Why error shrinks

Each week, more games become **known outcomes** rather than **predicted probabilities**. The unknown tail shrinks, and so does projection error. Note that even late-season error does not reach zero because the final game is still unknown at each team's last pre-game projection.

Chicago Bears: the model updates its belief.



Reading the chart

Pre-season prior

~7.6 projected wins, driven by 2024 form.

Blended strength leans on prior season.

As wins accumulate

Projection rises steadily.

Current-season evidence overtakes the prior.

Convergence

Projection converges near actual 11 wins.

Dashed line = actual final wins. By late season, projection ≈ truth.

A complete pipeline, **honest and reproducible.**

- 01 End-to-end pipeline**
Raw scrape → cleaning → features → model → projection.
- 02 Built on real PFR data**
No synthetic or proxy inputs: auditable, reproducible.
- 03 Leakage-safe by construction**
All rolling features use only prior games; time-based split.
- 04 Handles ties correctly**
Counted as 0.5 in projections, removed for binary fit.
- 05 Handles 2022 canceled game**
Bills vs. Bengals never replayed; pipeline treats it explicitly.
- 06 Validation-based selection**
Best 2024 validation log loss decided the final model.
- 07 Final model stays interpretable**
Logistic coefficients map to football-meaningful drivers.
- 08 Produces the exact deliverable**
544-row CSV of projected records, before every 2025 game.

No injuries, no QB changes, no EPA.

Football context the model does not see

Personnel Injuries Star players in/out can swing a win probability by 10+ points.	Personnel QB changes Starting QB identity is the single biggest predictor this model could miss.	Personnel Roster moves Trades, signings, suspensions, IR designations.	Game context Weather Wind, cold, precipitation reshape passing efficiency.
Game context Rest days & travel Short weeks, time zones, long road trips.	Market signal Betting lines Spreads and totals aggregate enormous prior information.	Granularity Player-level data No individual production, snap counts, or matchup data.	Granularity EPA / play-by-play Modern team-strength estimates ride on EPA, I used none.

Implication

The largest future gains likely come from **richer football context**, not from swapping logistic regression for a deeper algorithm.

Where the next yards come from.

Tier 1 · Quick wins

cheap data, immediate gain

Rest-day & short-week flags

Thursday games, bye-week boost, mini-bye.

Turnover differential entering game

Already in raw data, just need to engineer the feature.

Division-game indicator

Familiarity tightens point spreads.

Recent-form weighted stats

Last 4 games vs. season-long average.

Tier 2 · Meaningful adds

moderate effort, real signal

Starting QB identity

Single largest missing predictor.

Probability calibration

Isotonic / Platt scaling on validation slice.

More historical seasons

Carefully, must respect 17- vs 16-game regimes.

Richer team efficiency stats

3rd-down %, red-zone %, pace, etc.

Tier 3 · Ambitious

heavy lift, biggest ceiling

EPA / play-by-play features

Modern team-strength gold standard.

Injury reports

Active / inactive lists, snap-weighted impact.

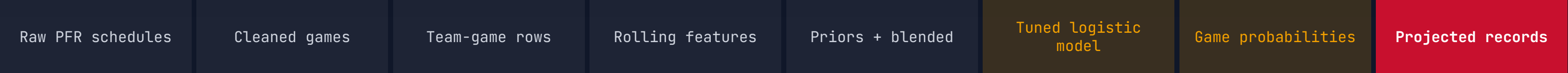
Schedule-strength adjustments

Iterated Elo-style adjustment.

Bayesian / hierarchical model

Native uncertainty bands on projected wins.

The pipeline, end to end



In one sentence

The final product is beyond a simple game predictor, it is a **rolling, season-long** record projection engine.

Input 1,359 games 2021–2025 · Pro Football Reference	Model Tuned logistic Validation AUC 0.711 · 2025 AUC 0.654	Deliverable 544 projections every team · before every 2025 game
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